
name: StudyG Learning Platform

overview: Build a full-stack virtual group study platform using React (Vite), Convex backend, BetterAuth authentication, and AI-powered quiz generation. The app enables users to create learning sessions with uploaded materials, track progress in real-time, collaborate with others via chat and shared annotations, take AI-generated quizzes, and manage revision with spaced repetition.

todos:

- id: phase-1-scaffold
content: "Phase 1: Create Vite + React project, install Convex + BetterAuth + Tailwind + Shadcn/ui, configure auth, build layout shell with routing (login, register, dashboard, session pages)"
status: pending
- id: phase-2-sessions
content: "Phase 2: Build Convex schema for courses/sessions/materials, implement session creation form, file upload with Convex storage, material page viewer component"
status: pending
- id: phase-3-progress
content: "Phase 3: Implement progress tracking mutations/queries, learning speed controls, estimated completion time, scheduled check-in functions with UI notifications"
status: pending
- id: phase-4-quizzes
content: "Phase 4: Build AI quiz generation via Convex actions + OpenAI, quiz taking UI with MCQ/short answer, scoring, and performance analysis"
status: pending
- id: phase-5-group
content: "Phase 5: Add group study features - invite codes, session member sidebar with real-time presence, chat system, shared annotations on materials"
status: pending
- id: phase-6-revision
content: "Phase 6: Implement spaced repetition revision system, dashboard with learning stats/charts, session history page, revision queue with priority sorting"
status: pending
isProject: false

StudyG – Virtual Group Learning Companion

Tech Stack

- **Frontend:** React 19 + Vite + TypeScript
- **Styling:** Tailwind CSS v4 + Shadcn/ui
- **Routing:** React Router v7
- **Backend:** Convex (real-time database, file storage, scheduled functions, actions)
- **Auth:** BetterAuth via [@convex-dev/better-auth](https://github.com/convex-dev/better-auth)
- **AI:** OpenAI API (via Convex actions) for quiz generation and material processing

Architecture Overview

```
graph TB
    subgraph frontend [React SPA - Vite]
        Pages[Pages and Components]
        ConvexClient[Convex React Client]
        AuthClient[BetterAuth Client]
    end

    subgraph convexBackend [Convex Backend]
        Queries[Queries - Real-time]
        Mutations[Mutations]
        Actions[Actions - AI Integration]
        Scheduler[Scheduled Functions]
        FileStorage[File Storage]
        BetterAuthComp[BetterAuth Component]
    end

    subgraph external [External Services]
        OpenAI[OpenAI API]
    end

    Pages --> ConvexClient
    Pages --> AuthClient
    ConvexClient -->|"useQuery / useMutation"| Queries
    ConvexClient --> Mutations
    AuthClient --> BetterAuthComp
    Mutations --> Scheduler
    Actions --> OpenAI
    Mutations --> FileStorage
```

Convex Database Schema

Key tables (defined in `convex/schema.ts`):

- **courses** - subjects/courses created by users
- **chapters** - chapters within courses
- **studySessions** - core entity: learning period, scope, settings, invite code
- **sessionMembers** - users in a session with role, status, personal learning speed
- **materials** - uploaded docs/images linked to sessions (references `_storage`)
- **progress** - per-user, per-material page tracking with timestamps
- **checkIns** - periodic check-in prompts and responses during sessions
- **quizzes** - AI-generated quizzes with questions, options, explanations
- **quizAttempts** - user answers and scores per quiz
- **chatMessages** - real-time group chat within sessions
- **annotations** - shared notes on specific pages of materials
- **revisionItems** - topics flagged for review using spaced repetition scheduling
- **learningStats** - aggregated per-user statistics (streak, hours, avg score)

erDiagram

```
studySessions ||--o{ sessionMembers : has
studySessions ||--o{ materials : contains
studySessions ||--o{ chatMessages : has
studySessions ||--o{ quizzes : generates
studySessions ||--o{ checkIns : schedules
materials ||--o{ annotations : has
materials ||--o{ progress : tracks
quizzes ||--o{ quizAttempts : has
courses ||--o{ chapters : contains
courses ||--o{ studySessions : organizes
```

Key Feature Implementation Details

1. Authentication (BetterAuth + Convex)

- Install `@convex-dev/better-auth` and `better-auth`
- Register the BetterAuth component in `convex/convex.config.ts`
- Configure auth in `convex/auth.config.ts` with email/password + social providers
- Create `src/lib/auth-client.ts` for the frontend BetterAuth client
- Wrap app with `ConvexProvider` + auth context

2. Study Session Creation

- Form to set: title, course, chapters/topics, start/end time, learning speed (min/page)
- Upload multiple documents/images using Convex file storage (`generateUploadUrl` pattern)
- Option to make session public or generate a private invite code for group study
- On creation, schedule periodic check-in functions via `ctx.scheduler.runAt()`

3. Learning Speed and Progress Tracking

- Default speed set at session creation (e.g., 3 min/page), adjustable per member
- Material viewer shows current page with navigation
- Progress auto-saves on page change via mutation
- Estimated completion time calculated from speed x remaining pages
- Real-time progress bar visible to all session members via Convex subscriptions

4. Periodic Check-Ins (Convex Scheduled Functions)

- When session starts, schedule check-in mutations at intervals (e.g., every 20 min)
- Each check-in triggers a UI notification asking "How's your progress?"
- User responds: on-track / struggling / ahead, with optional notes
- Responses stored and used to adjust revision priority

5. AI-Generated Quizzes (Convex Actions + OpenAI)

- After a session completes (or on-demand), trigger a Convex action
- Action extracts text from uploaded materials (OCR for images via OpenAI Vision, text extraction for docs)
- Sends content to OpenAI with prompt to generate quiz questions (MCQ + short answer)
- Questions tagged by topic and difficulty level
- Results stored in `quizzes` table, user attempts in `quizAttempts`
- Score analysis identifies weak areas -> feeds into revision system

6. Revision System (Spaced Repetition)

- After each quiz, topics where user scored poorly are flagged as `revisionItems`
- Uses a simplified SM-2 algorithm to schedule next revision date
- Dashboard shows upcoming revisions sorted by urgency
- Revision sessions re-surface materials and re-quiz on weak topics
- Critical items (repeatedly failed) get highlighted attention

7. Group Study (Real-Time Collaboration)

- **Join:** via invite code or public session browser
- **Progress sync:** all members' current page and status visible in a sidebar (Convex `useQuery` subscriptions)
- **Chat:** real-time text chat within the session (simple `chatMessages` table with live query)
- **Shared annotations:** members can add notes/highlights on specific pages, visible to all in real-time
- **Presence:** member status (active/idle/away) updated via heartbeat mutation

8. Session History and Dashboard

- All past sessions listed with stats (duration, pages covered, quiz scores)
- Aggregate stats: total hours, streak, average performance
- Visual charts for learning trends over time
- Quick access to revision queue

Project Structure

```
StudyG/
  src/
    components/
      ui/                # Shadcn components
      auth/              # Login, Register forms
      session/           # Session creation, viewer, progress
      materials/         # Upload, page viewer
      quiz/              # Quiz taking, results
      chat/              # Group chat
      annotations/       # Shared notes overlay
      dashboard/         # Stats, history, revision queue
      layout/            # Navbar, sidebar, layout wrappers
    lib/
      auth-client.ts     # BetterAuth client config
      utils.ts           # Shared utilities
      spaced-rep.ts       # SM-2 algorithm implementation
    pages/
      HomePage.tsx
      DashboardPage.tsx
      SessionCreatePage.tsx
      SessionActivePage.tsx
      QuizPage.tsx
      RevisionPage.tsx
```

```
    SessionHistoryPage.tsx
App.tsx
main.tsx
convex/
  schema.ts          # Full database schema
  convex.config.ts   # Component registration
  auth.config.ts     # BetterAuth configuration
  auth.ts            # Auth helpers
  sessions.ts        # Study session CRUD + queries
  materials.ts       # File upload + management
  progress.ts        # Progress tracking mutations/queries
  checkIns.ts        # Scheduled check-in functions
  quizzes.ts         # Quiz generation actions + storage
  chat.ts            # Chat message mutations/queries
  annotations.ts     # Annotation mutations/queries
  revision.ts        # Revision system logic
  stats.ts           # Learning stats aggregation
  crons.ts           # Recurring jobs (stats, cleanup)
public/
index.html
vite.config.ts
tailwind.config.ts
package.json
tsconfig.json
```

Implementation Phases

The work is divided into 6 phases, each building on the previous:

Phase 1 - Project scaffolding, Convex setup, BetterAuth integration, basic UI shell

Phase 2 - Study session CRUD, material uploads, page viewer

Phase 3 - Progress tracking, learning speed, check-in system

Phase 4 - AI quiz generation, quiz taking UI, scoring

Phase 5 - Group study: real-time sync, chat, shared annotations

Phase 6 - Revision system, dashboard stats, session history, polish